ARRAYS

It is collection of homogeneous [same type] variables.

Array is nothing but collection of contiguous memory locations, where we can store and manage more than one value of same type under one name.

It is a derived data type.

It is an implicit / internal pointer.

It is a implicit const pointer

It is one of data structure.

Advantages:

Generally to store several values of same type, we have to declare several variables. Here we have to remember all these variable names also. When the program is too big, it is very difficult to remember all the variable names. In this situation, the only solution is array.

Array reduce program length.

Array minimize the errors.

In functions to carry several values of same type at a time, we are using arrays.

It allows to arrange our data in a order.

Disadvantage:

Array size is Constant Positive Integer value. Due to this we are not able to change the array size at run time. Sometimes it causes memory wastage / shortage.

In C language we are using

1. **One dimensional arrays**
2. **Multi dimensional arrays**

<u>One dimensional arrays:</u>

- **An array with one row and several columns.**
- **An array with single subscripting operator [ ] is called one dimensional array.**
- **It is an implicit single pointer.**

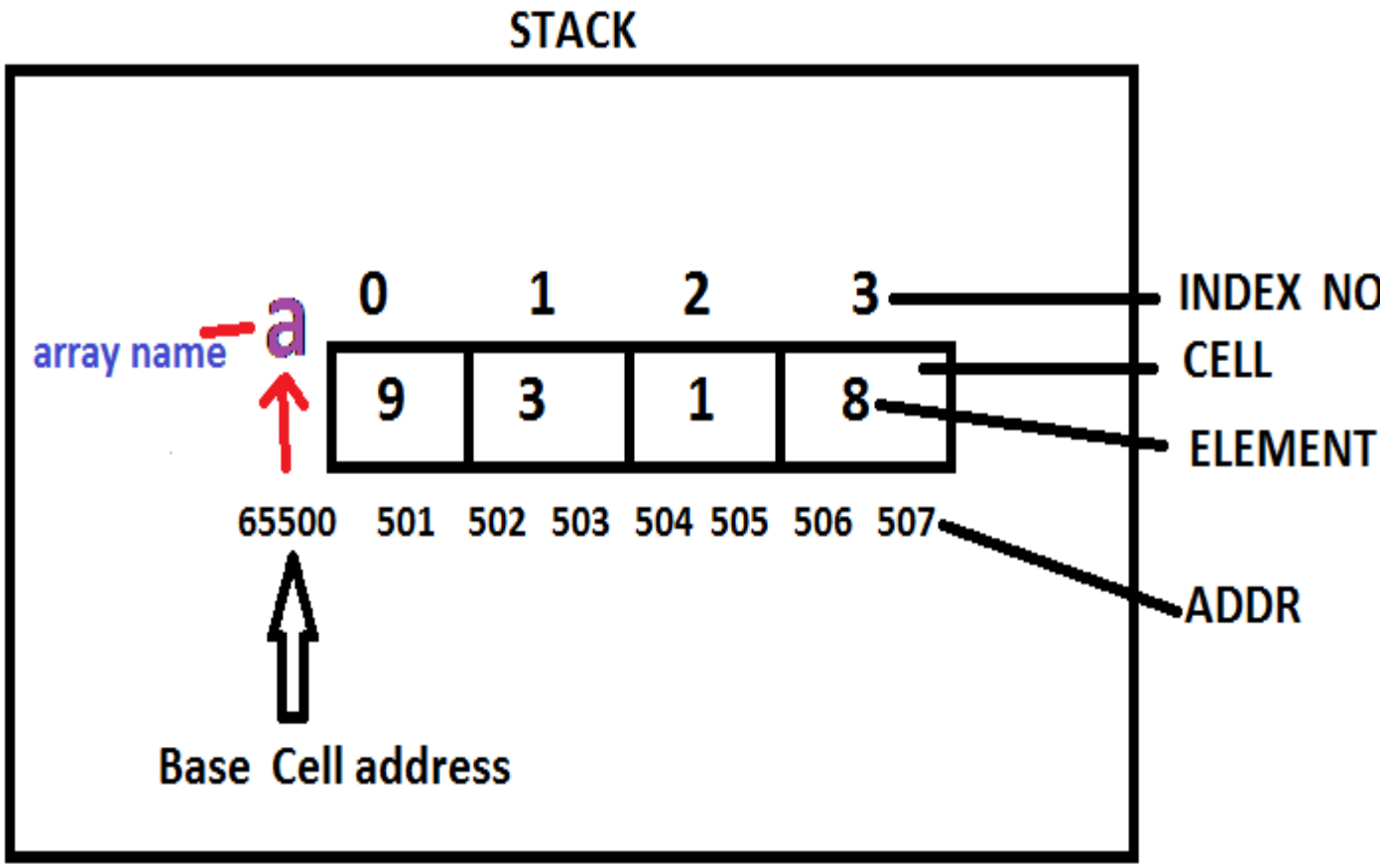
<u>Syntax</u>:

datatype variable[size] = {elements};

Eg:

```
int  a[4] = { 9, 3, 1, 8 };
```

<u>Memory allocation for array:</u>



Array is implicit pointer because of array variable stores base cell [0 cell 1st byte] address. Hence array variable value and 0 cell address both are same.

Array declaration methods:

int a[3]; Ok

int a[ ]; No

int a[3]={1,2,3}; Ok

int a[ ]={1,2,3}; Ok

int a[0]={1,2,3}; Ok

int a[-5]; No

int a[5.5]; No

int n = 5, a[n]; No

int a[3]={10,20}; Ok

int a[3]={1, 2, 3, 4}; No

int a[0]; error

#define n 5 /* macro */

int a[n]; Ok

const int n=5, a[n]; No

int a[5>3]; ➔ int a[1]; Ok

int a[3<2]; ➔ int a[0]; No

int a[2+3]; ➔int a[5]; Ok

int a[5%3]; ➔ int a[2]; Ok

int a[5%5]; ➔ int a[0]; No

int a[1,2,3]; ➔ error

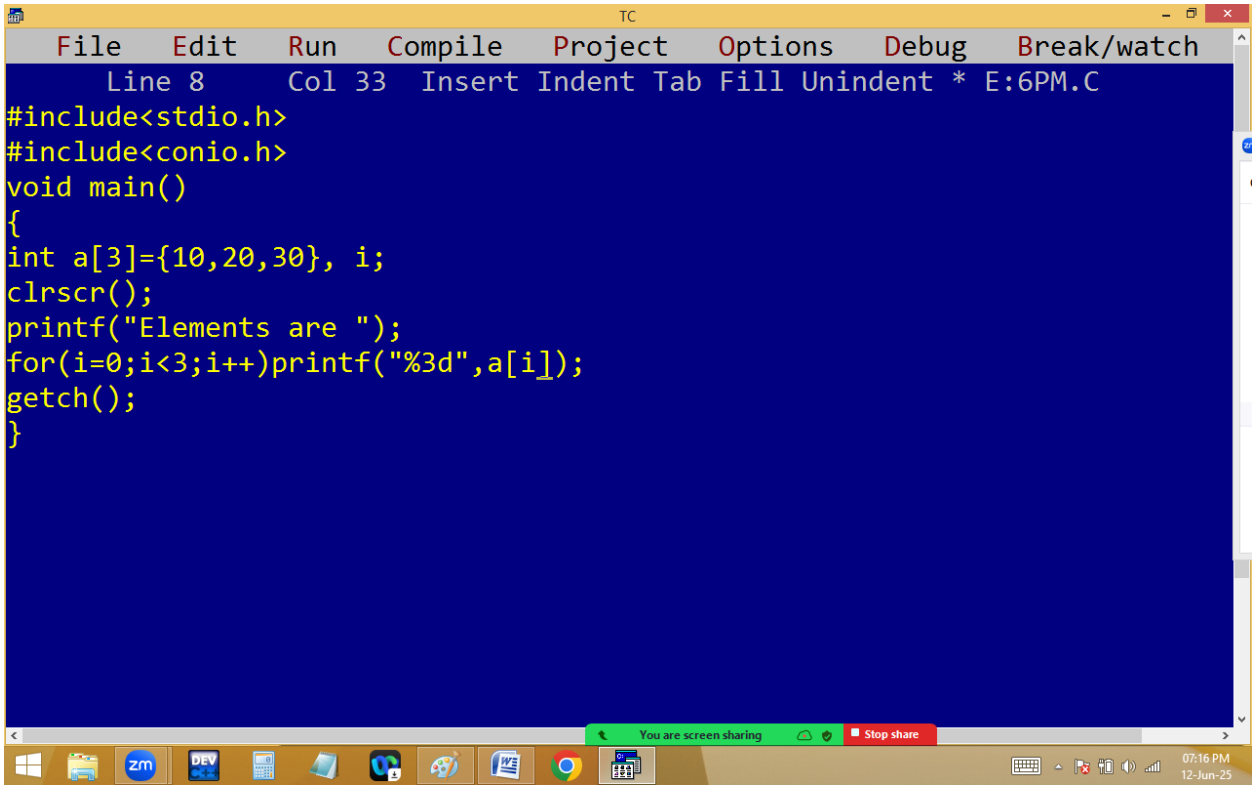
int a[40000]; ➔ 40000 * 2 = 80000 bytes ➔ No

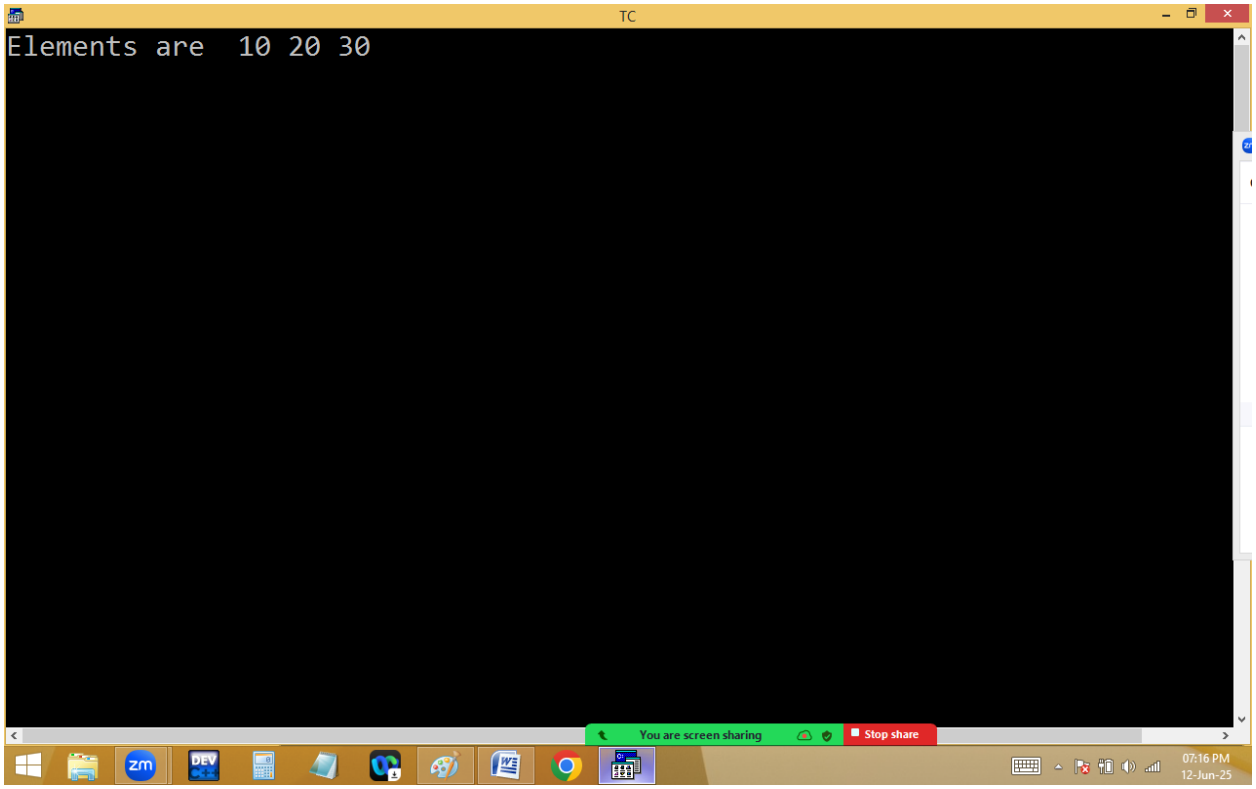
Note: Stack size is 65536 bytes(64kb) Only.

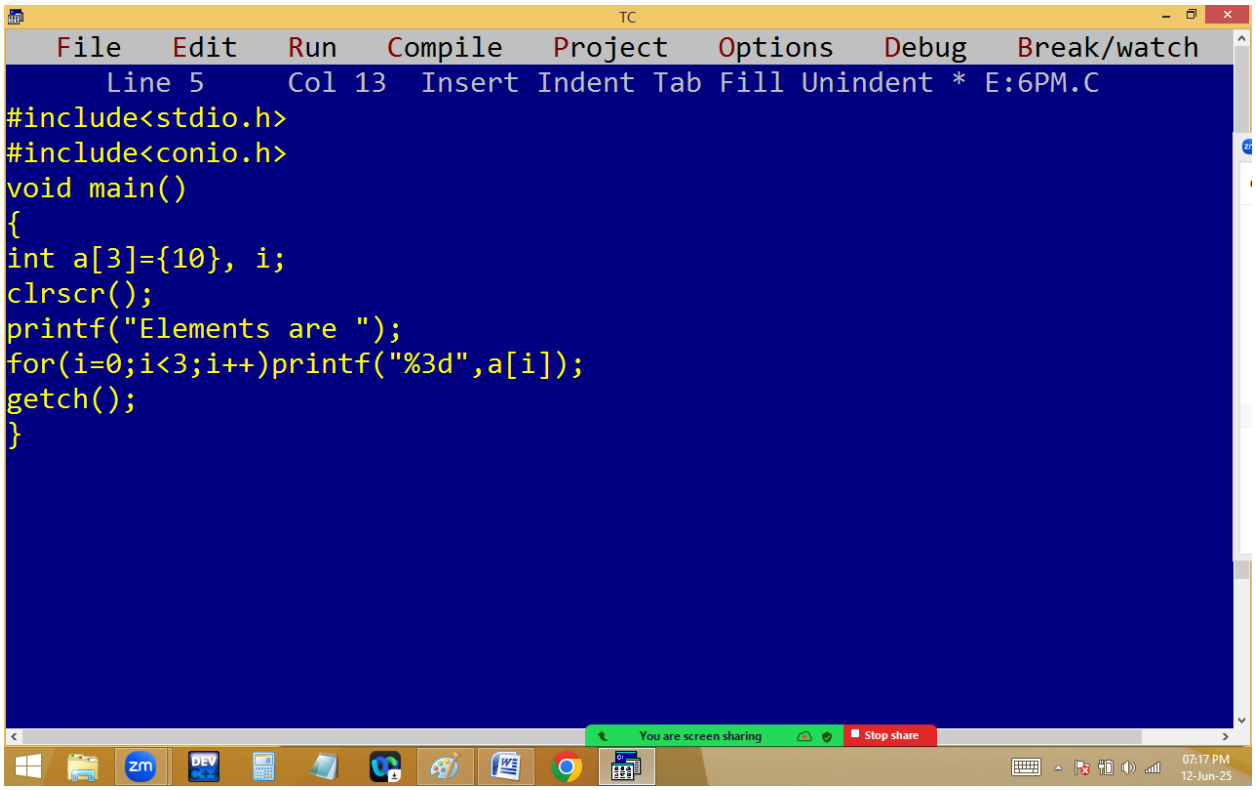
float a[10000]; Ok ➔ 10000 * 4 = 40000 bytes

float a[20000]; No ➔ 20000 * 4 = 80000 bytes

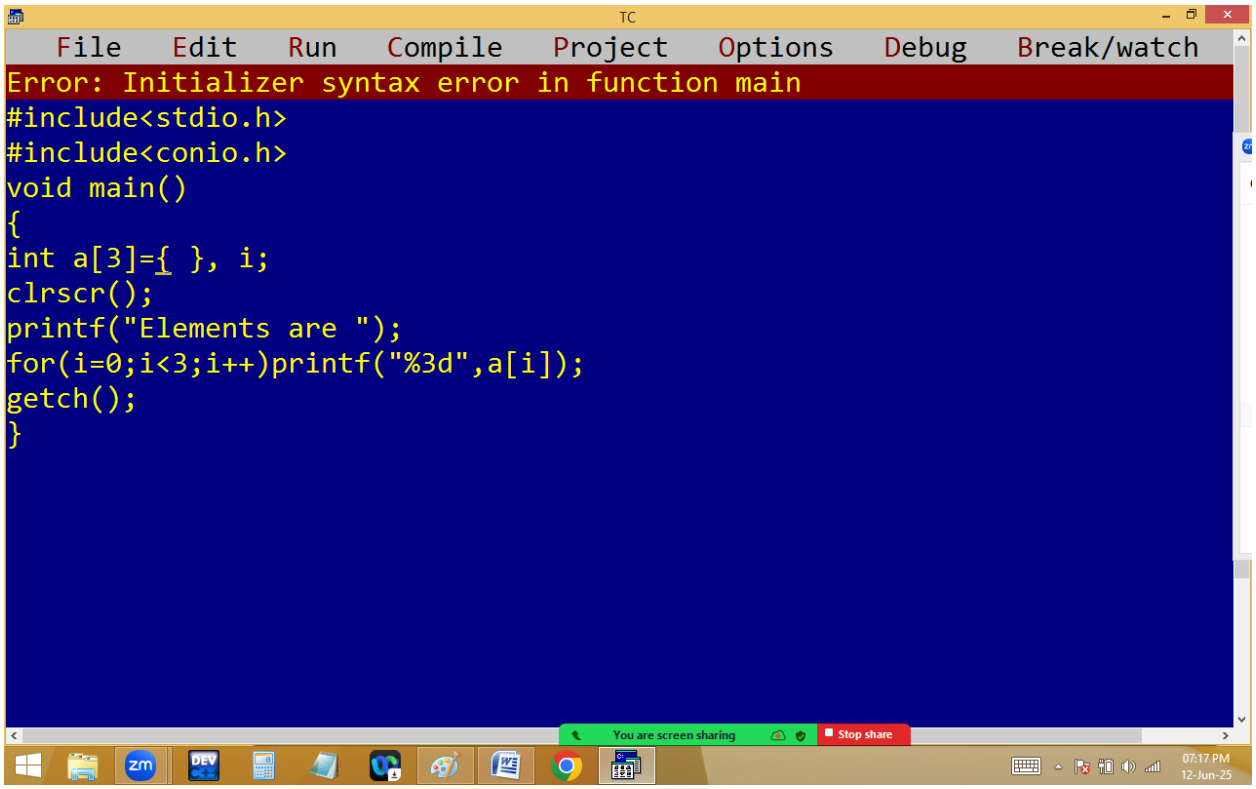
direct initialization of array elements:

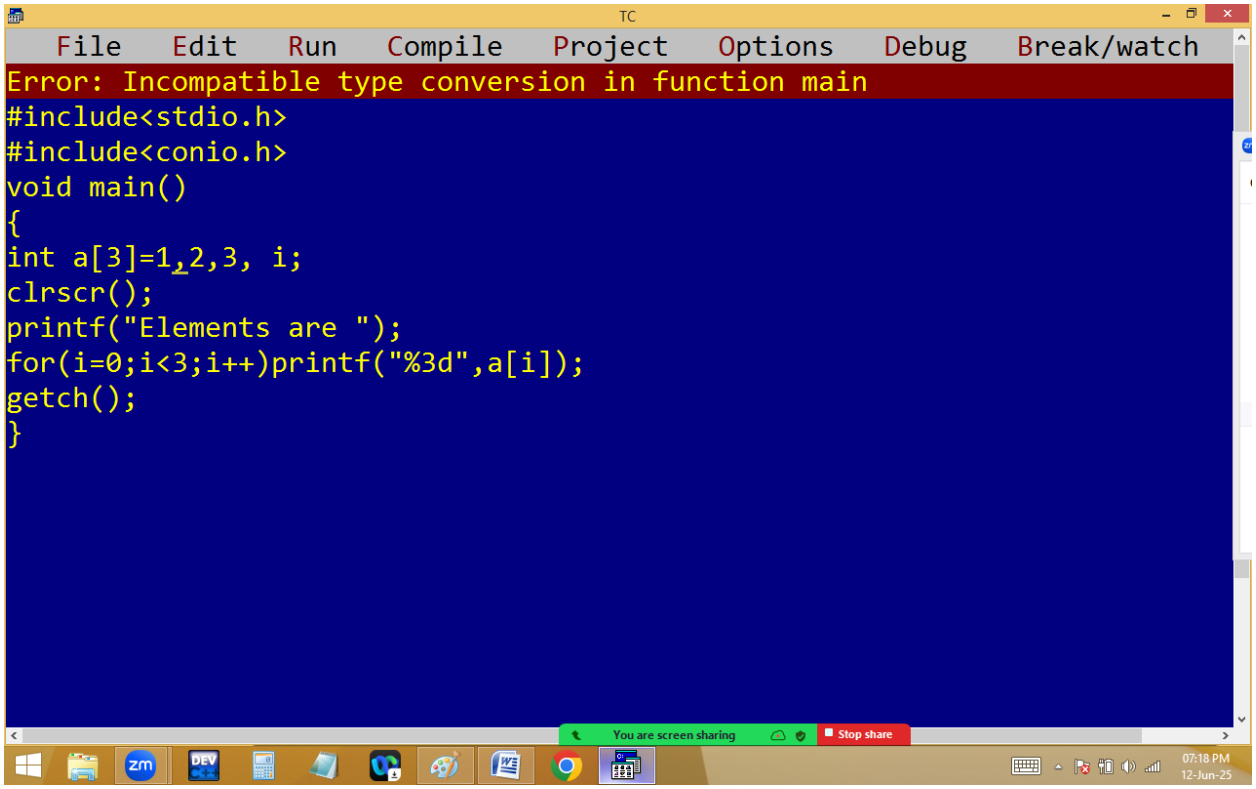


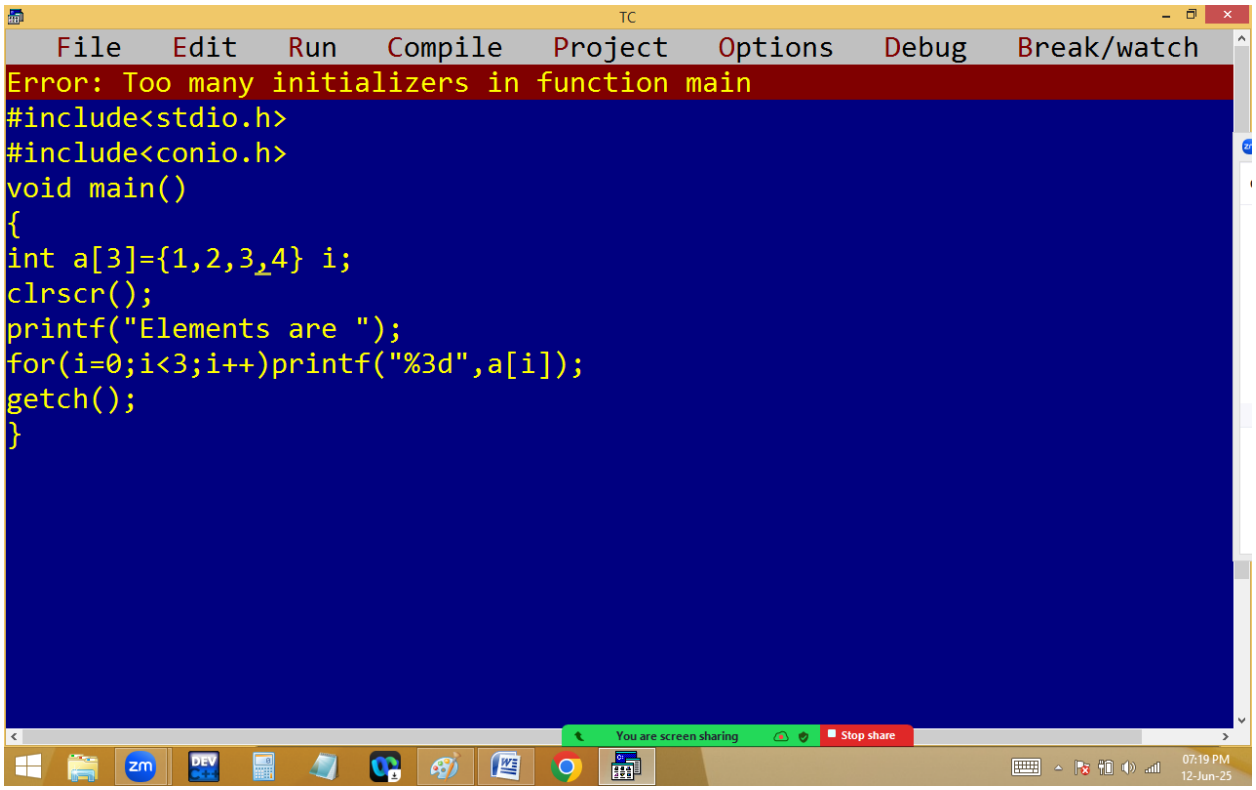


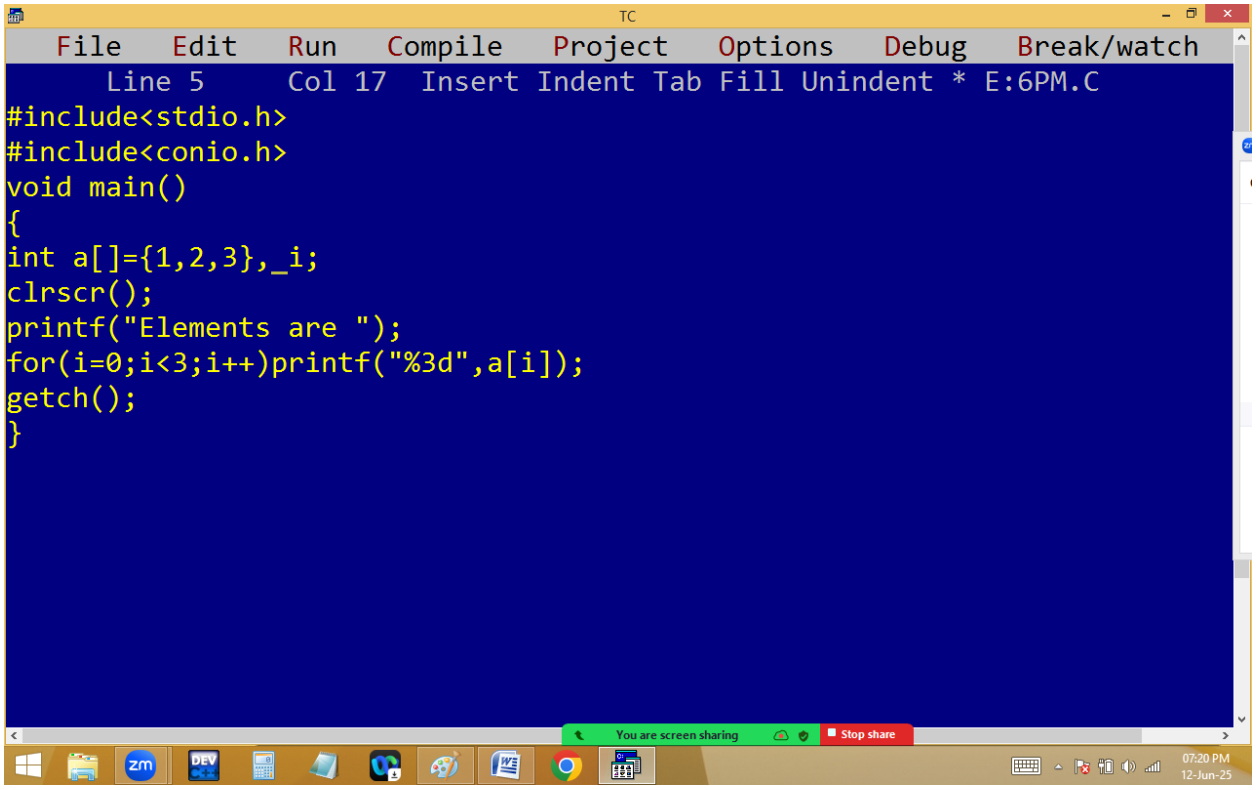

TC
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 13 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[3]={10}, i;
clrscr();
printf("Elements are ");
for(i=0;i<3;i++)printf("%3d",a[i]);
getch();
}
You are screen sharing
Stop share
07:17 PM
12-Jun-25

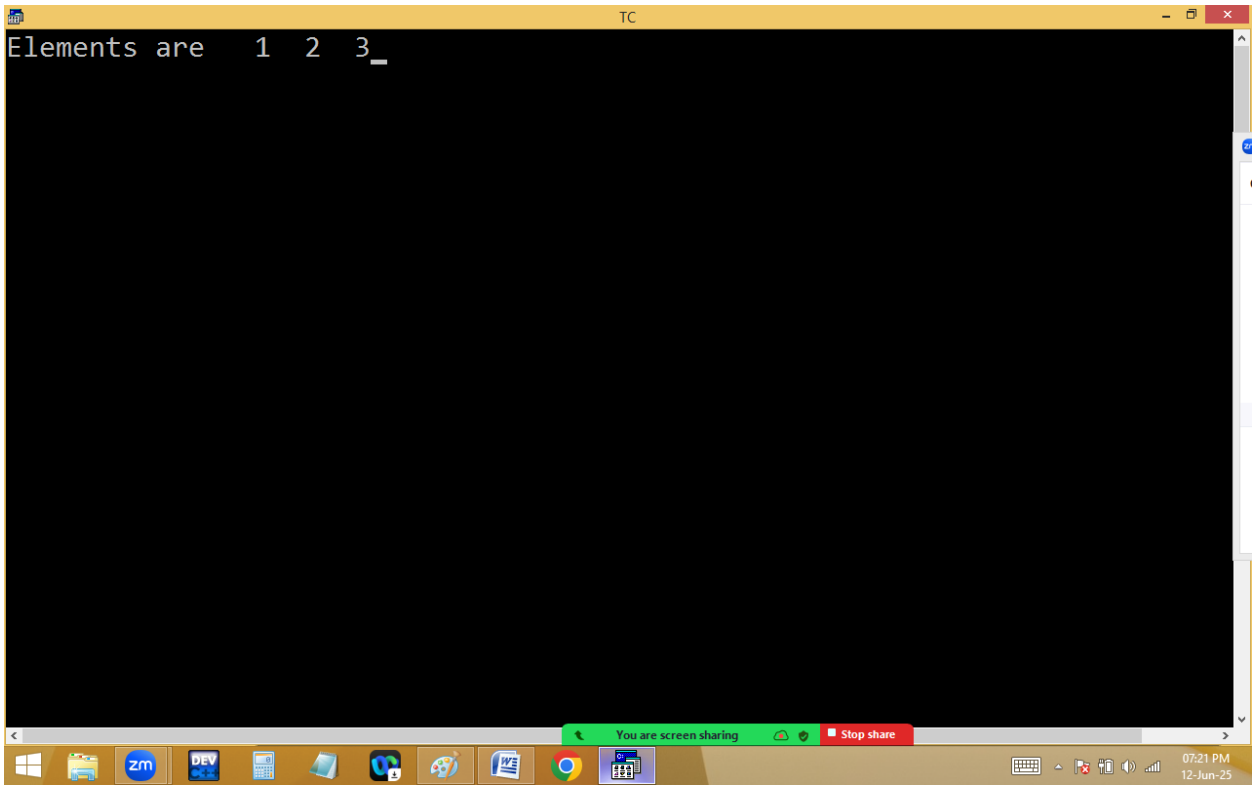

TC
Elements are 10 0 0
You are screen sharing
Stop share
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12-Jun-25

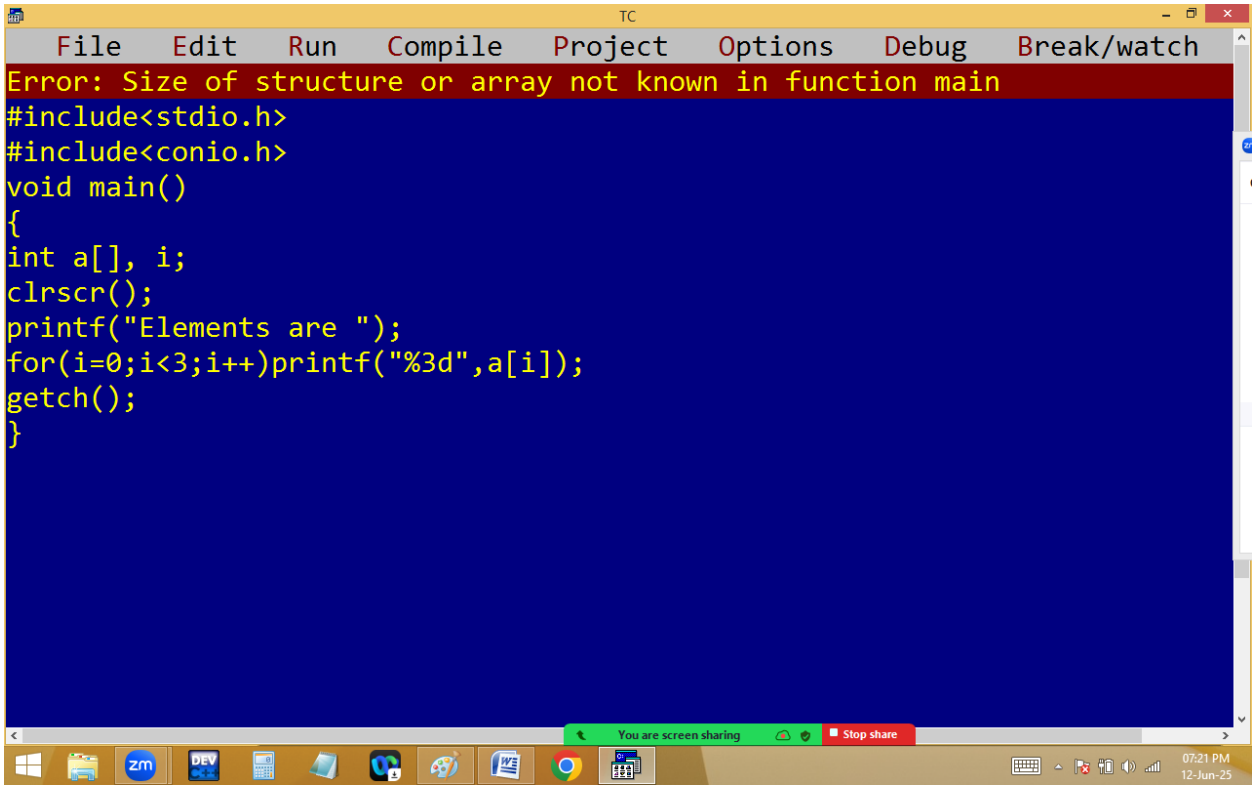

TC
File Edit Run Compile Project Options Debug Break/watch
Error: Initializer syntax error in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a[3]={ }, i;
clrscr();
printf("Elements are ");
for(i=0;i<3;i++)printf("%3d",a[i]);
getch();
}
You are screen sharing
Stop share
07:17 PM
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TC
File Edit Run Compile Project Options Debug Break/watch
Error: Incompatible type conversion in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a[3]=1,2,3, i;
clrscr();
printf("Elements are ");
for(i=0;i<3;i++)printf("%3d",a[i]);
getch();
}
You are screen sharing
Stop share
07:18 PM
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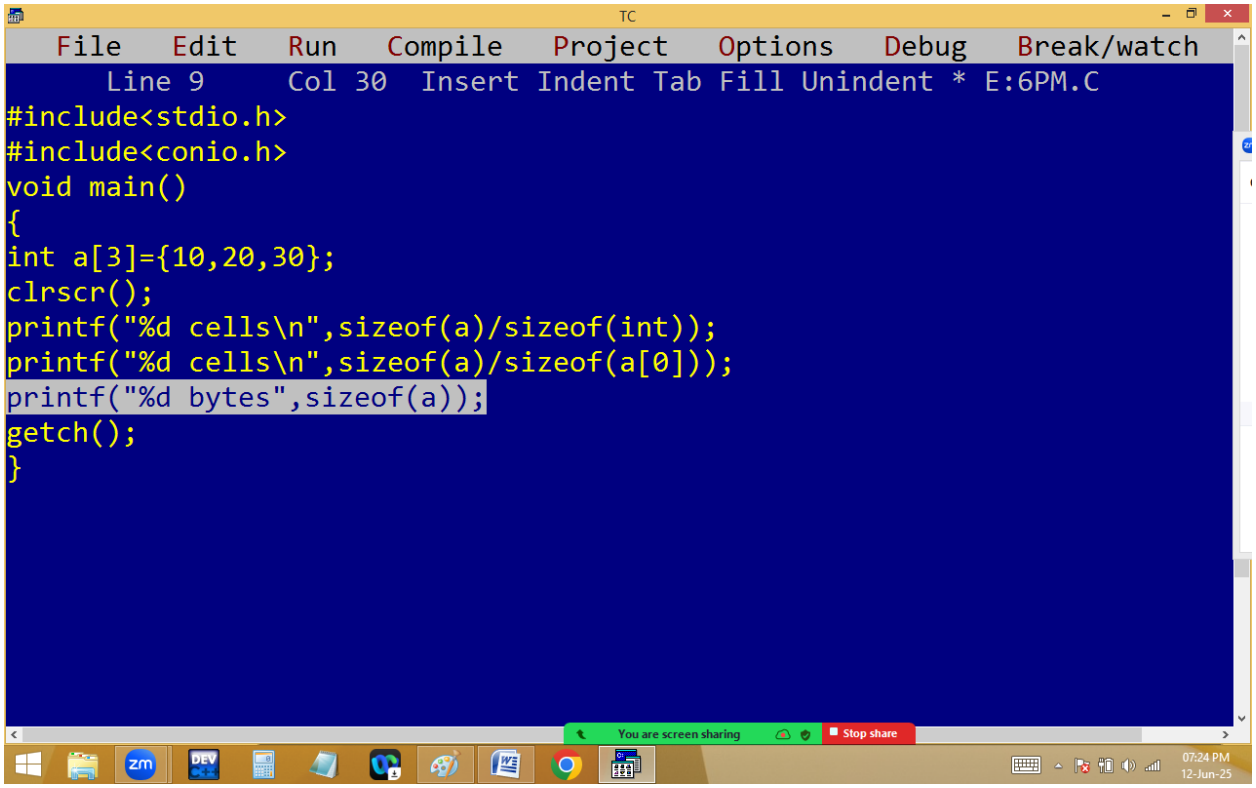

TC
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Error: Too many initializers in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a[3]={1,2,3,4} i;
clrscr();
printf("Elements are ");
for(i=0;i<3;i++)printf("%3d",a[i]);
getch();
}
You are screen sharing
Stop share
07:19 PM
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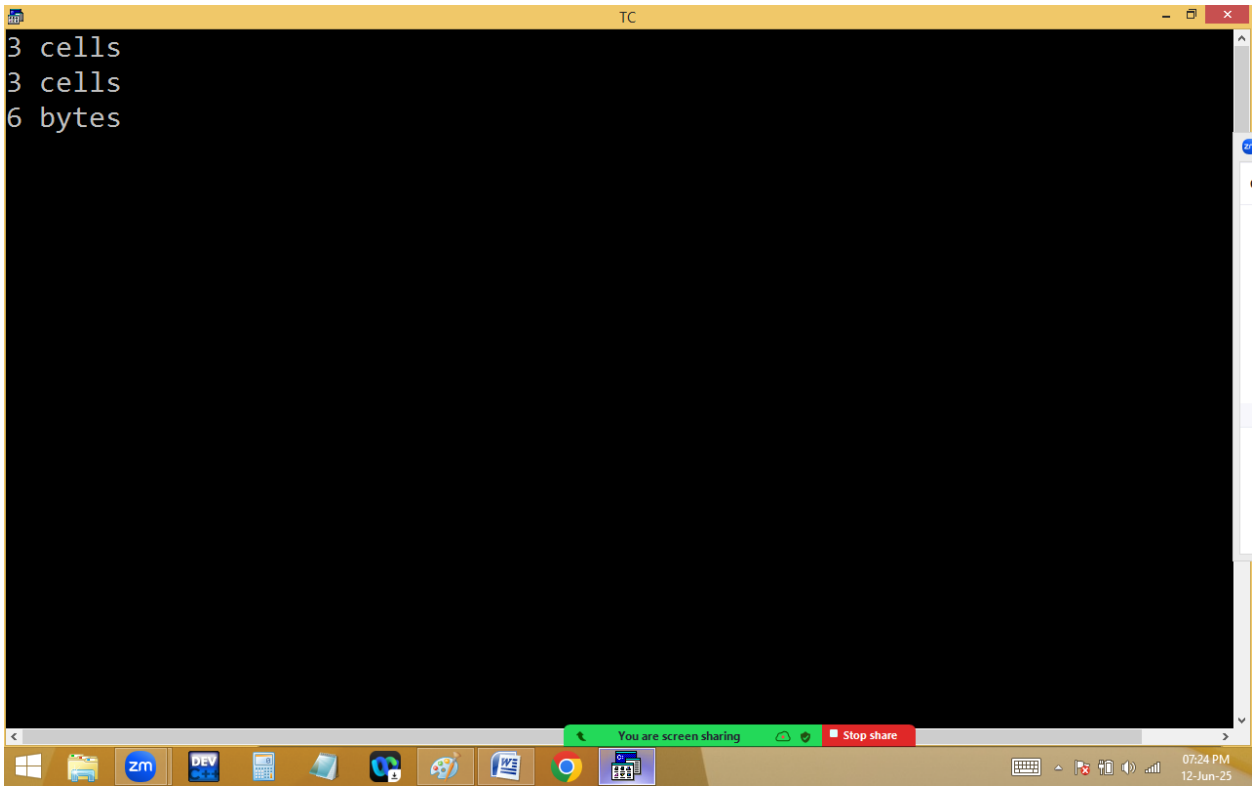

TC
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Line 5 Col 17 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[]={1,2,3},_i;
clrscr();
printf("Elements are ");
for(i=0;i<3;i++)printf("%3d",a[i]);
getch();
}
You are screen sharing
Stop share
07:20 PM
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TC
Elements are 1 2 3
You are screen sharing
Stop share
07:21 PM
12-Jun-25

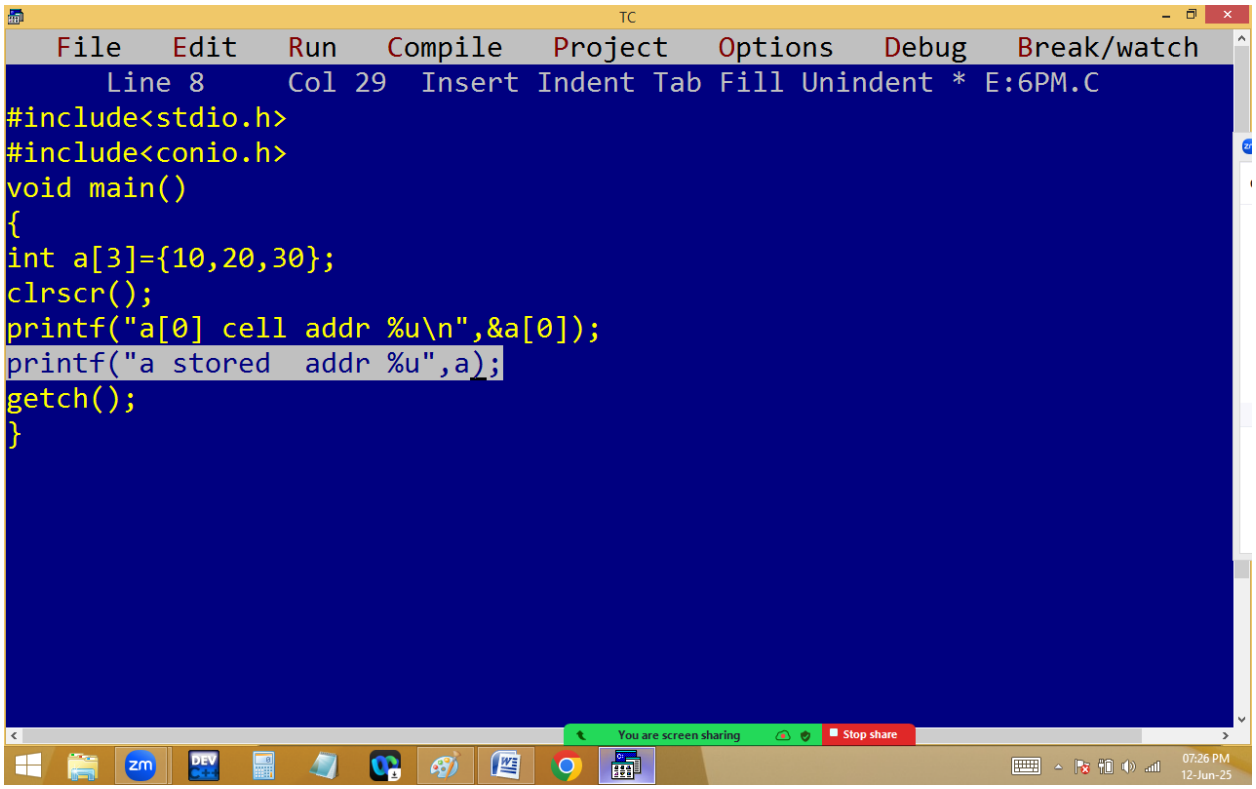

TC
File Edit Run Compile Project Options Debug Break/watch
Error: Size of structure or array not known in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a[], i;
clrscr();
printf("Elements are ");
for(i=0;i<3;i++)printf("%3d",a[i]);
getch();
}
You are screen sharing
Stop share
07:21 PM
12-Jun-25

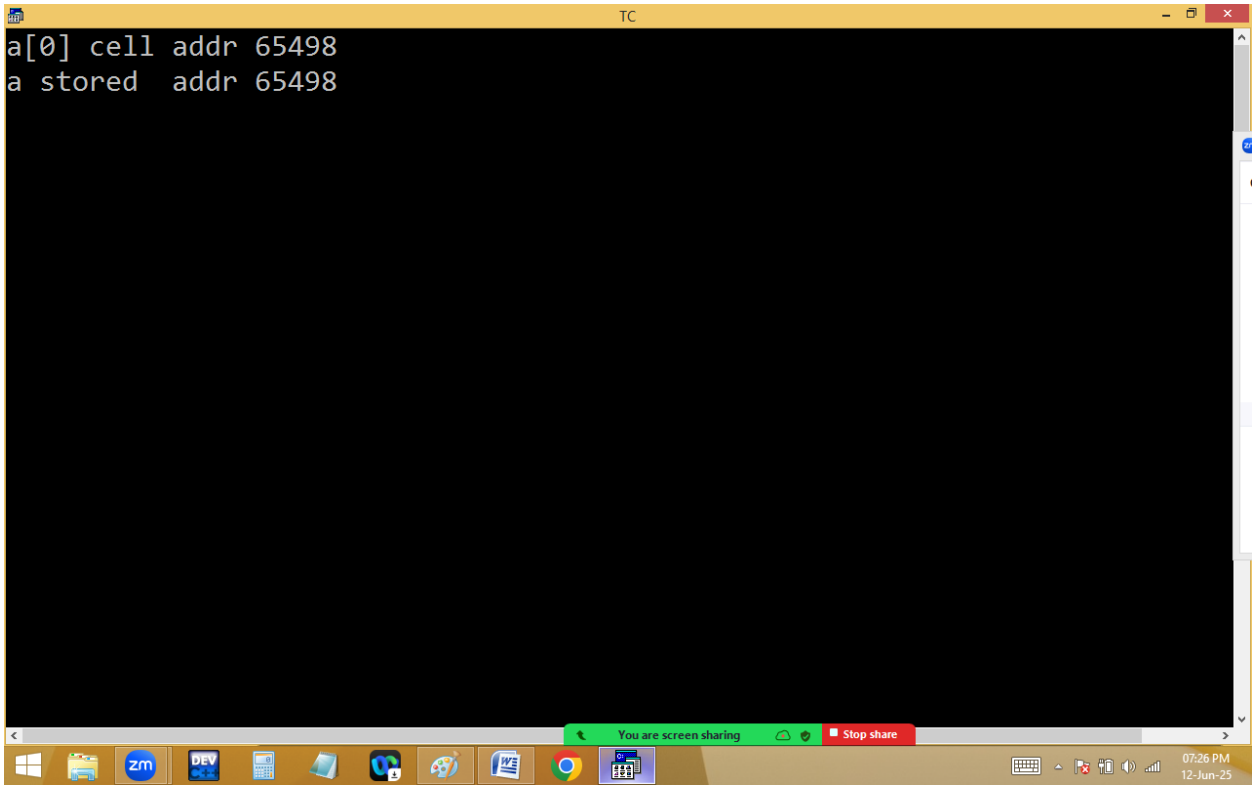
Finding array size:



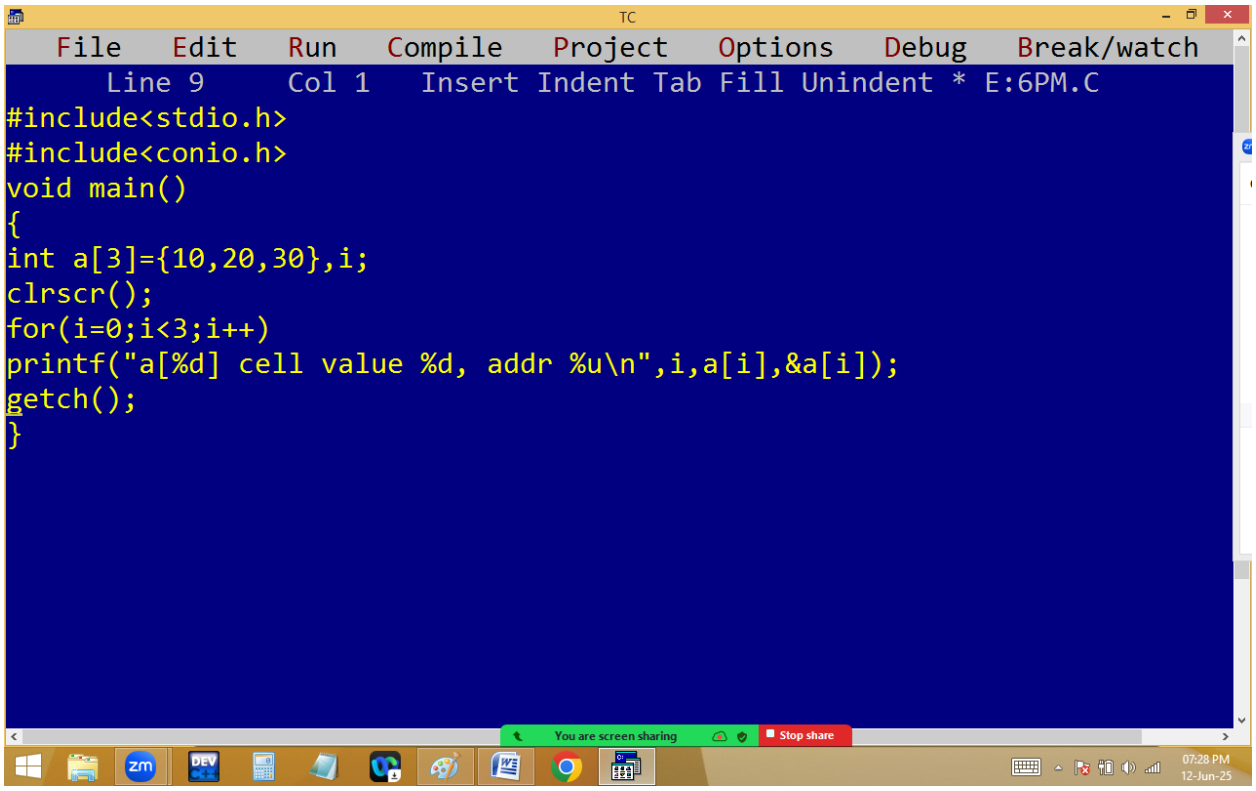


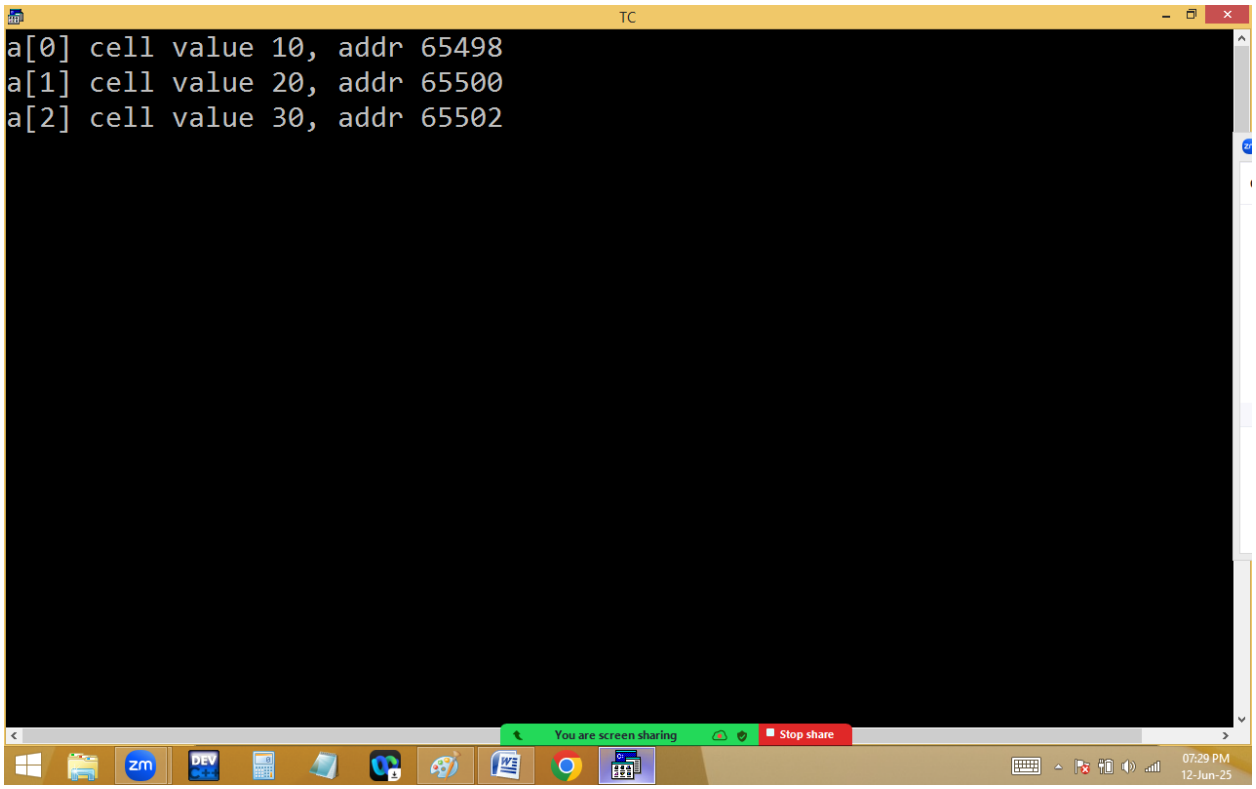
Finding array address:

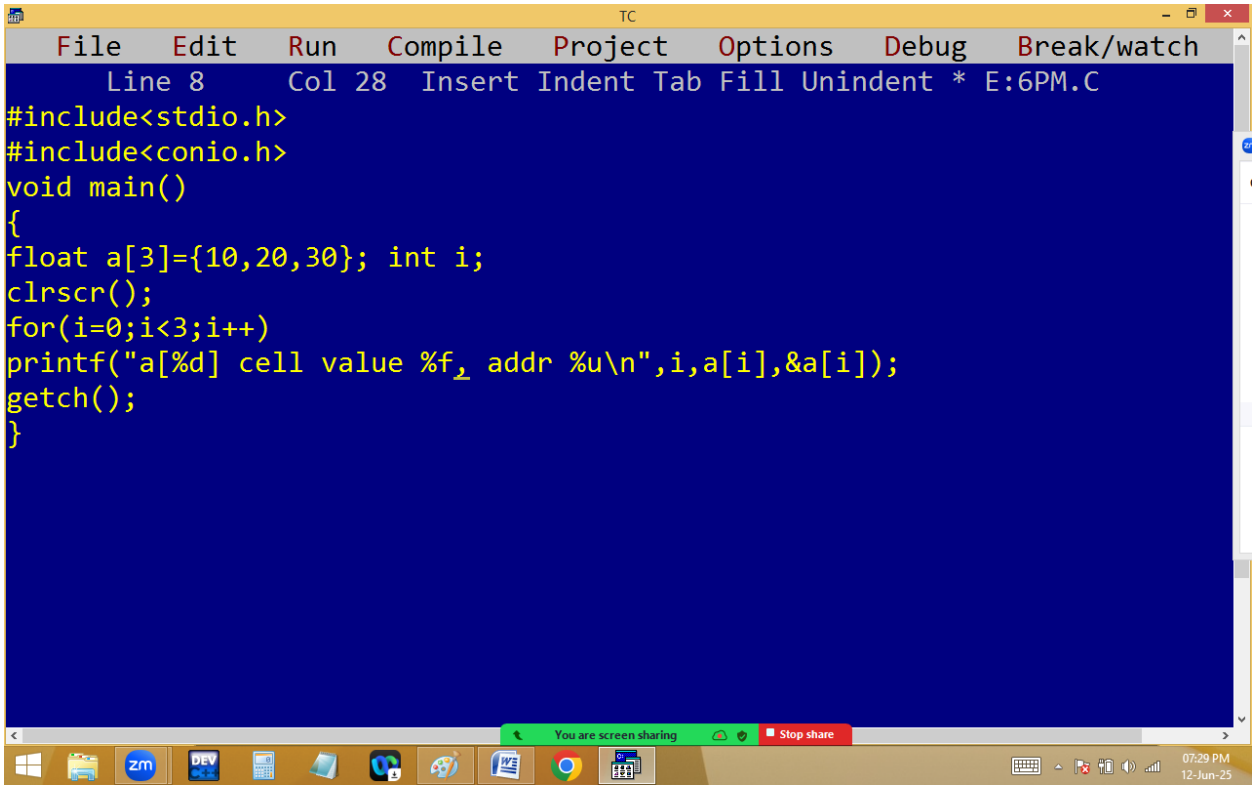

TC
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Line 8 Col 29 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[3]={10,20,30};
clrscr();
printf("a[0] cell addr %u\n",&a[0]);
printf("a stored addr %u",a);
getch();
}
You are screen sharing
Stop share
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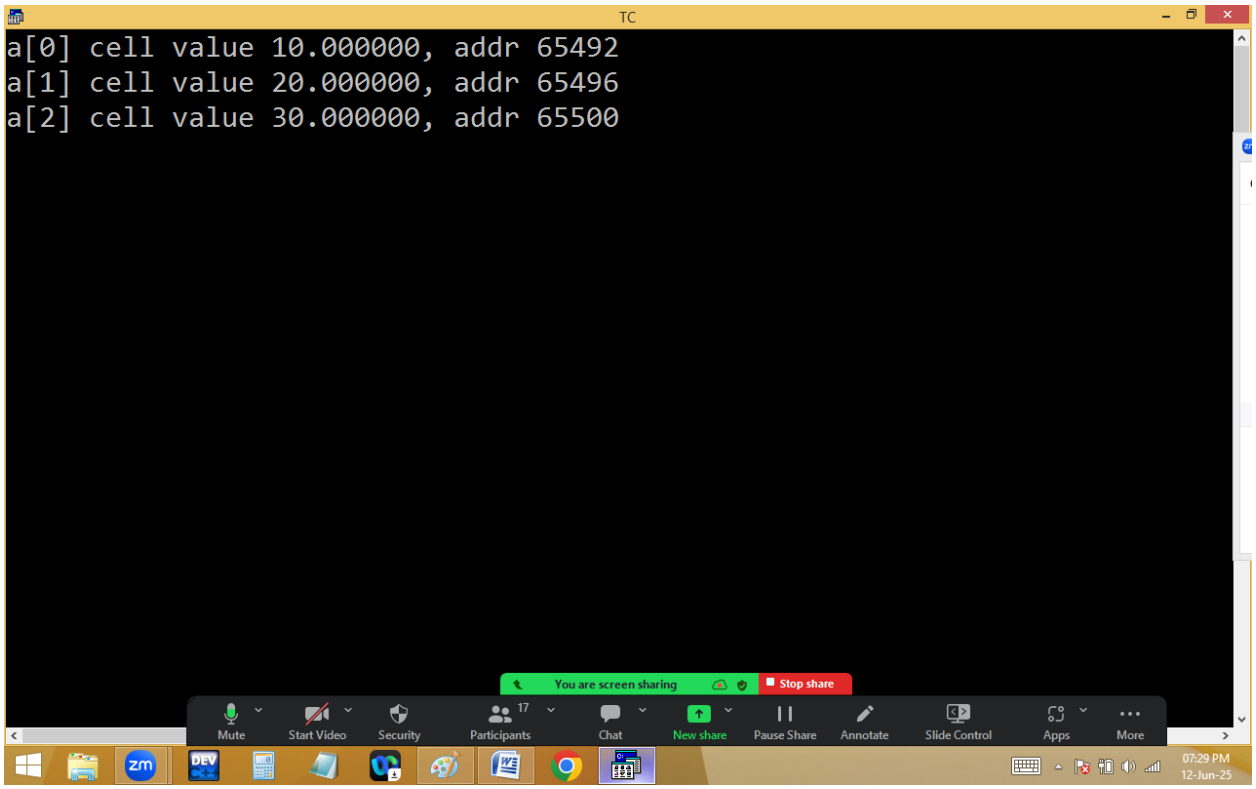

TC
a[0] cell addr 65498
a stored addr 65498
You are screen sharing
Stop share
07:26 PM
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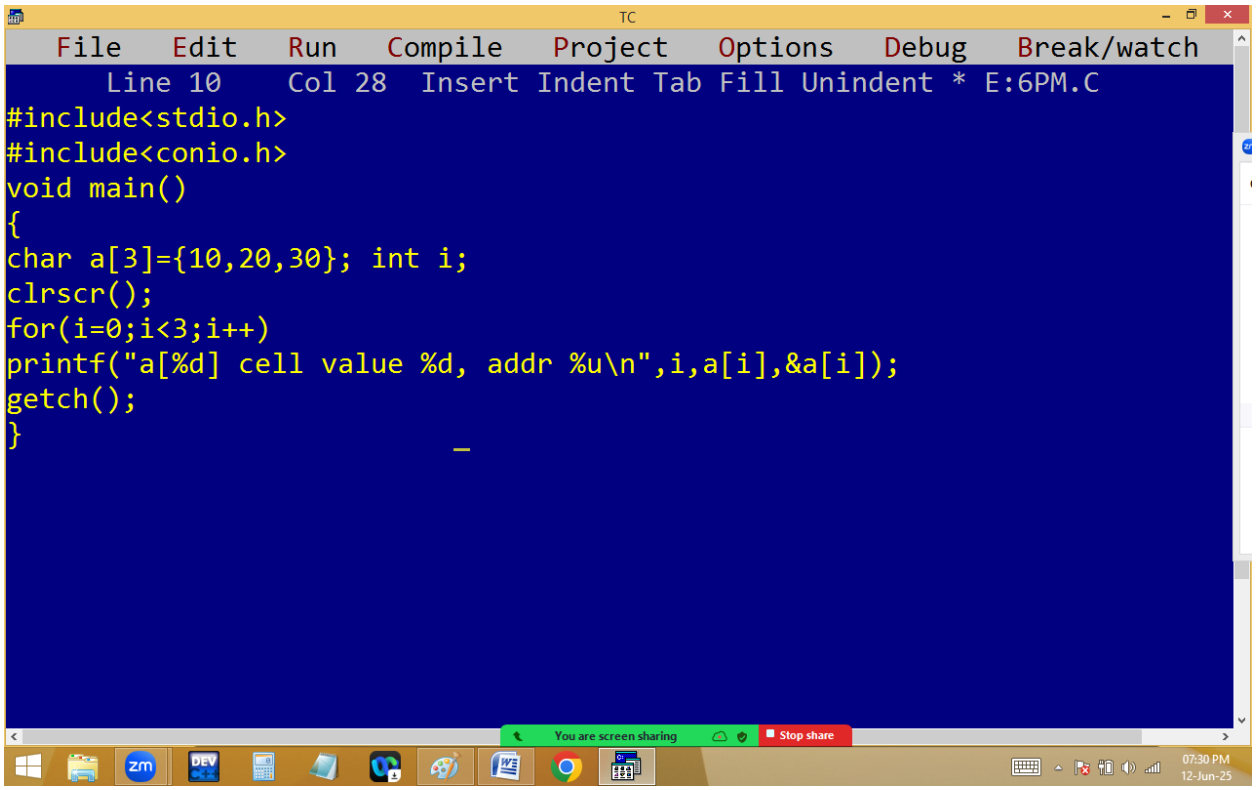
Finding cell no, value and address:

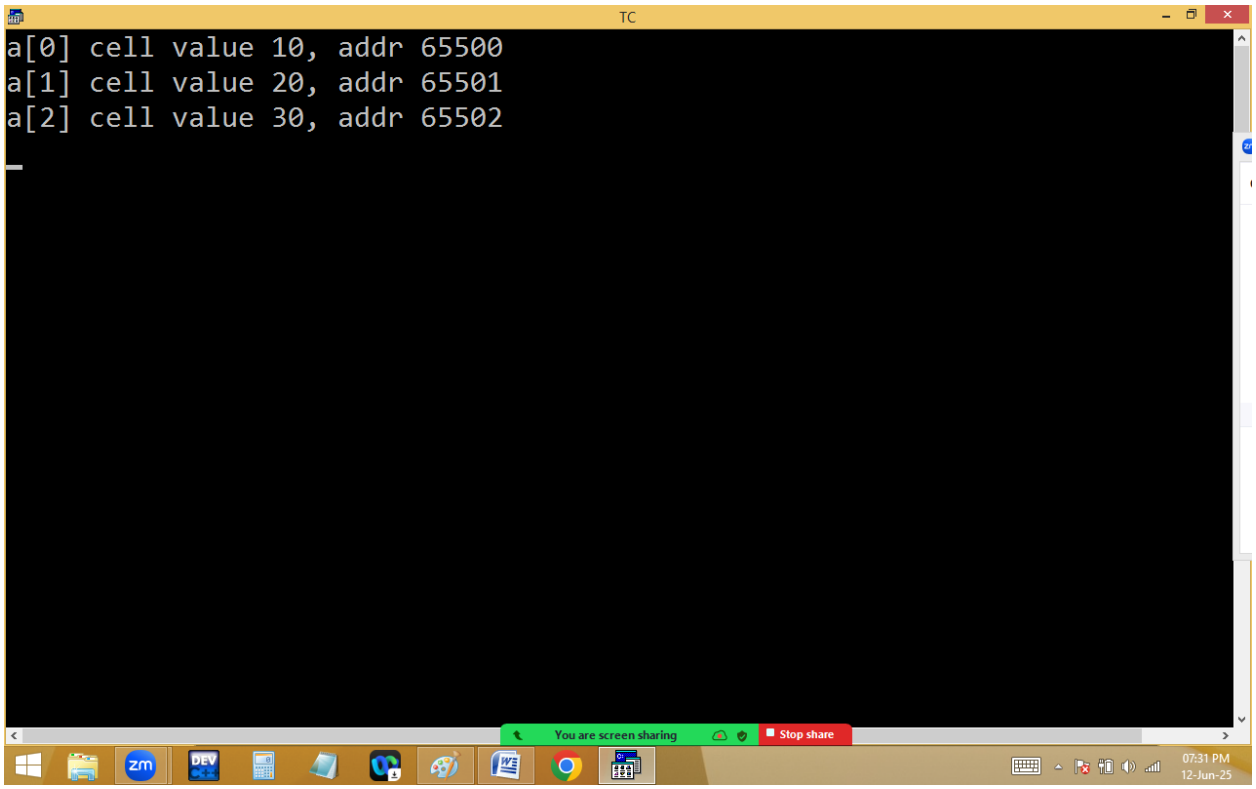


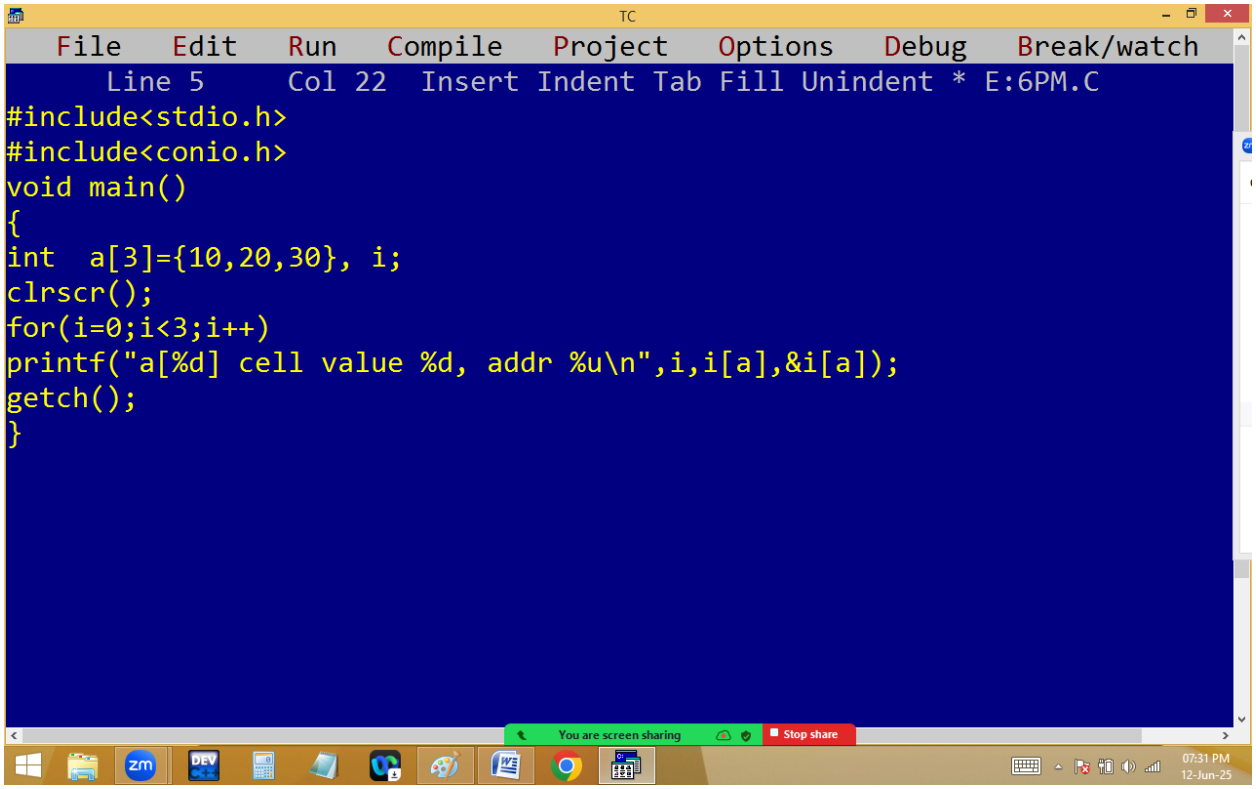

TC
a[0] cell value 10, addr 65498
a[1] cell value 20, addr 65500
a[2] cell value 30, addr 65502
You are screen sharing
Stop share
07:29 PM
12-Jun-25

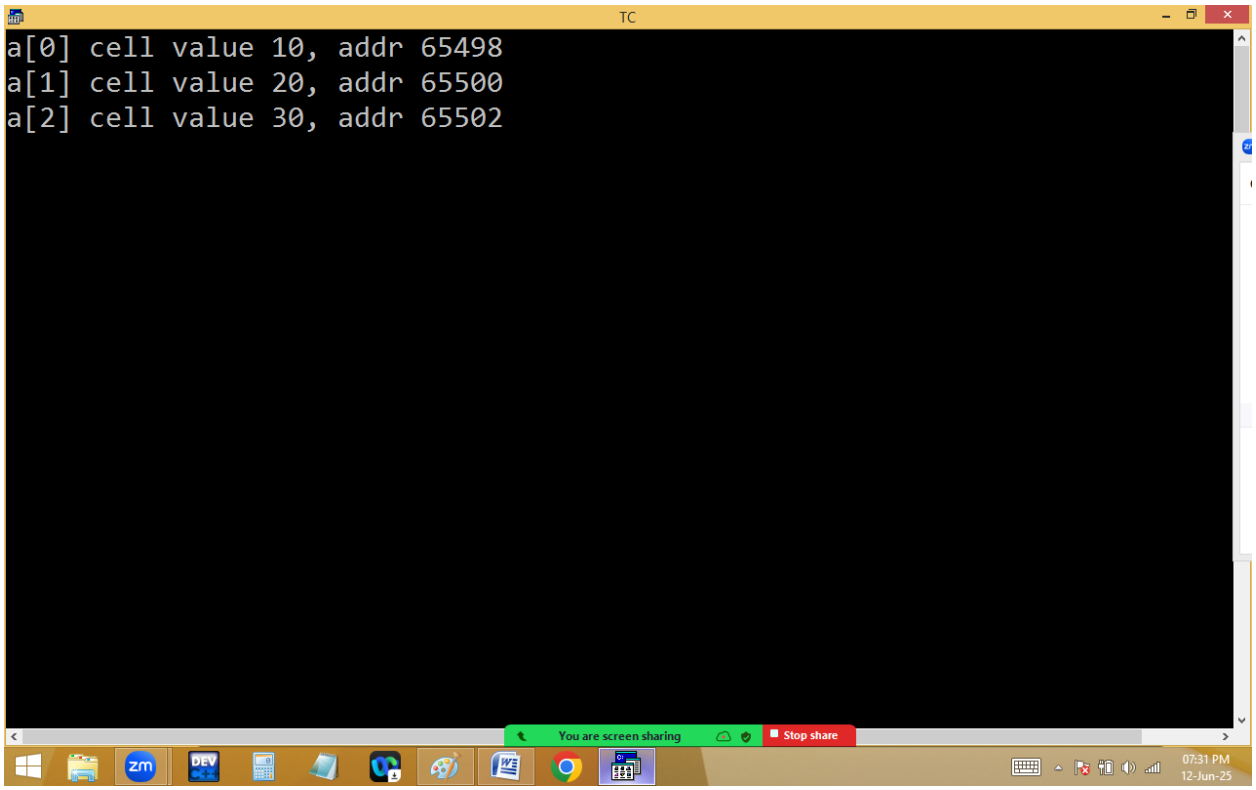

TC
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Line 8 Col 28 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
float a[3]={10,20,30}; int i;
clrscr();
for(i=0;i<3;i++)
printf("a[%d] cell value %f, addr %u\n",i,a[i],&a[i]);
getch();
}
You are screen sharing
Stop share
07:29 PM
12-Jun-25


TC
a[0] cell value 10.000000, addr 65492
a[1] cell value 20.000000, addr 65496
a[2] cell value 30.000000, addr 65500
You are screen sharing
Stop share
Mute
Start Video
Security
Participants
17
Chat
New share
Pause Share
Annotate
Slide Control
Apps
More
07:29 PM
12-Jun-25

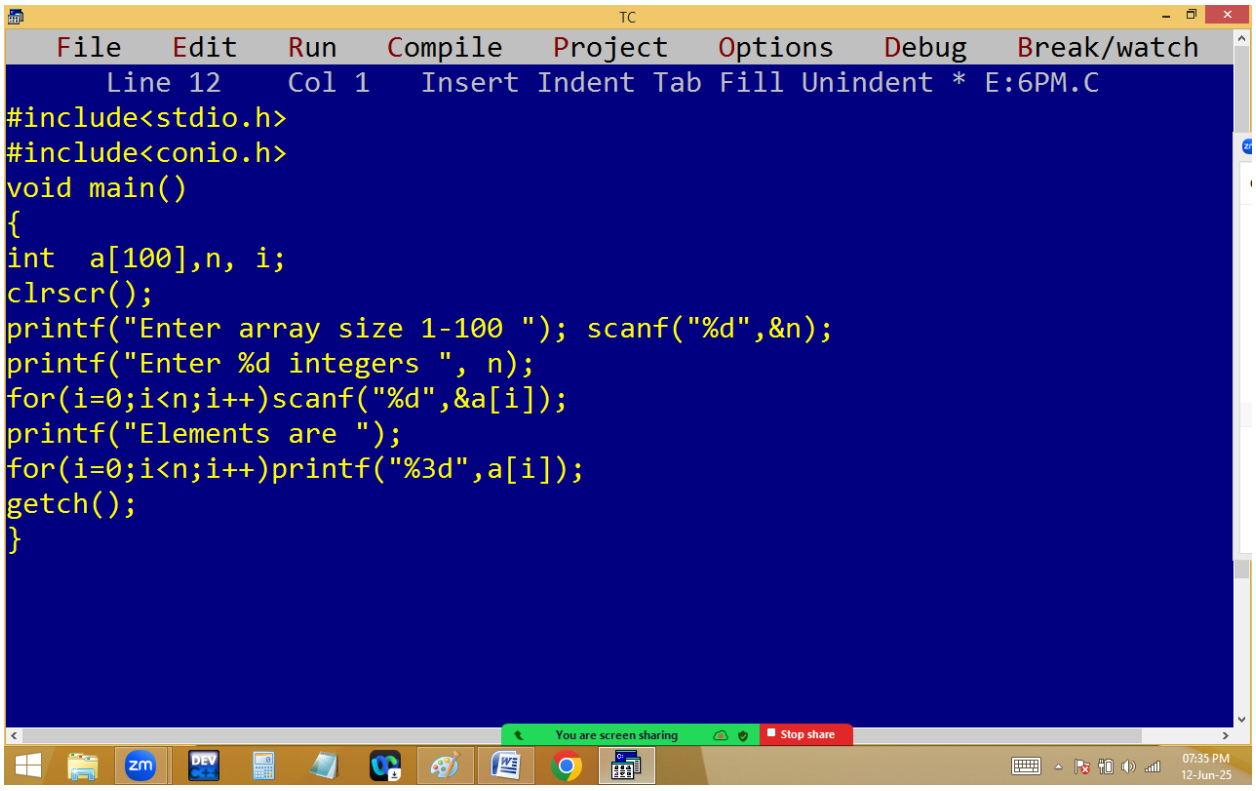

TC
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Line 10 Col 28 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char a[3]={10,20,30}; int i;
clrscr();
for(i=0;i<3;i++)
printf("a[%d] cell value %d, addr %u\n",i,a[i],&a[i]);
getch();
}
You are screen sharing
Stop share
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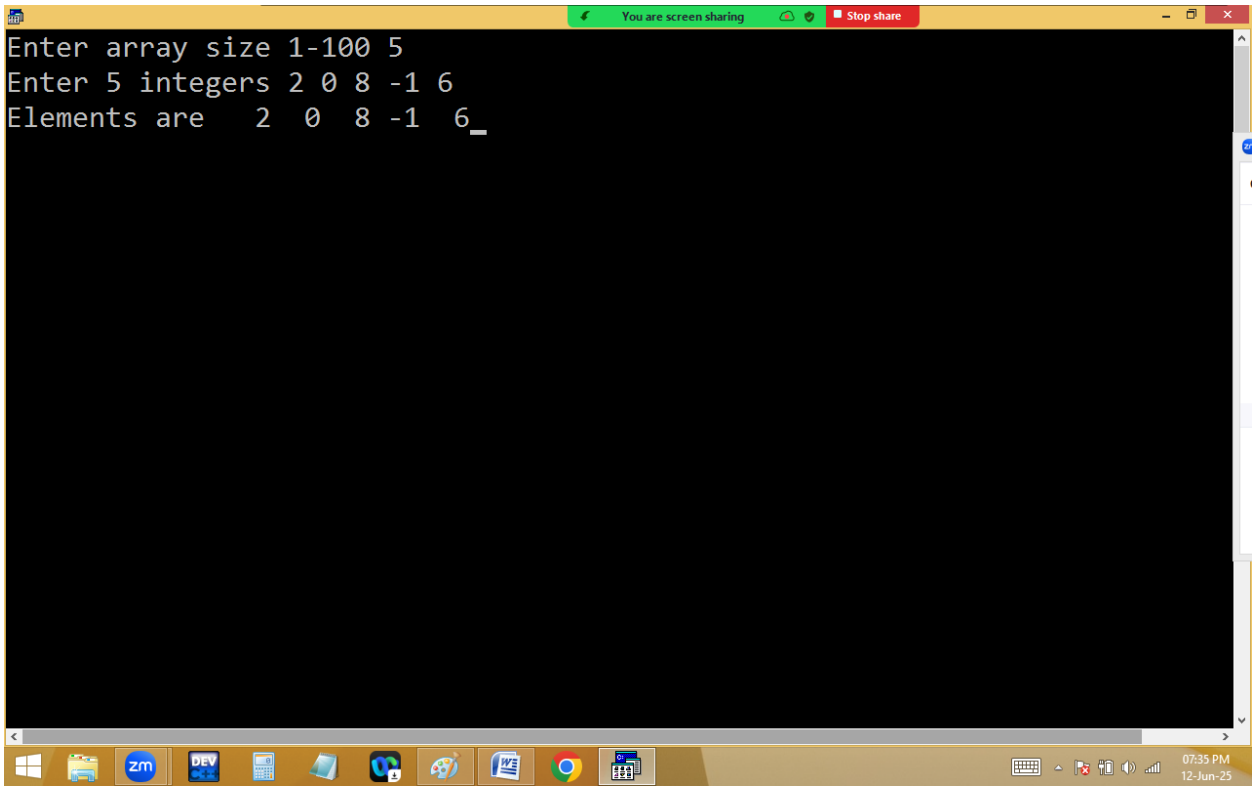

TC
a[0] cell value 10, addr 65500
a[1] cell value 20, addr 65501
a[2] cell value 30, addr 65502
You are screen sharing
Stop share
07:31 PM
12-Jun-25

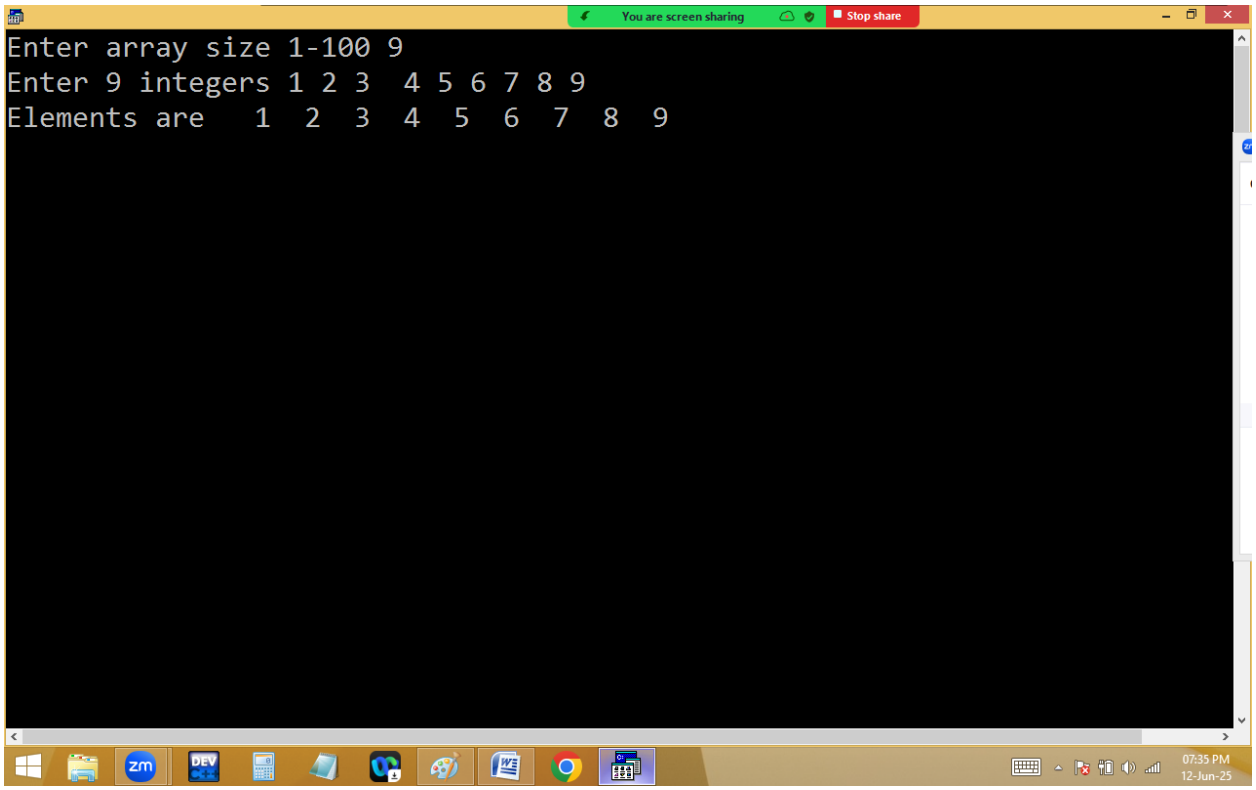

TC
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Line 5 Col 22 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[3]={10,20,30}, i;
clrscr();
for(i=0;i<3;i++)
printf("a[%d] cell value %d, addr %u\n",i,i[a],&i[a]);
getch();
}
You are screen sharing
Stop share
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Reading and printing n elements of array?


TC
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Line 12 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n, i;
clrscr();
printf("Enter array size 1-100 "); scanf("%d",&n);
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Elements are ");
for(i=0;i<n;i++)printf("%3d",a[i]);
getch();
}
You are screen sharing
Stop share
07:35 PM
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You are screen sharing
Stop share
Enter array size 1-100 5
Enter 5 integers 2 0 8 -1 6
Elements are 2 0 8 -1 6
07:35 PM
12-Jun-25


You are screen sharing
Stop share
Enter array size 1-100 9
Enter 9 integers 1 2 3 4 5 6 7 8 9
Elements are 1 2 3 4 5 6 7 8 9
07:35 PM
12-Jun-25